

Question 1 (Healthcare Analytics)

Program:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Product = c("P1","P2","P3","P4","P5","P6"),
```

```
  Sales = c(120,90,150,80,135,100),
```

```
  Profit = c(18,12,25,8,22,16),
```

```
  Category = c("Electronics","Clothing","Electronics","Furniture","Furniture","Clothing")
```

```
)
```

```
ggplot(data, aes(x = Sales, y = Profit, color = Category, size = Sales)) +
```

```
  geom_point(alpha = 0.8) +
```

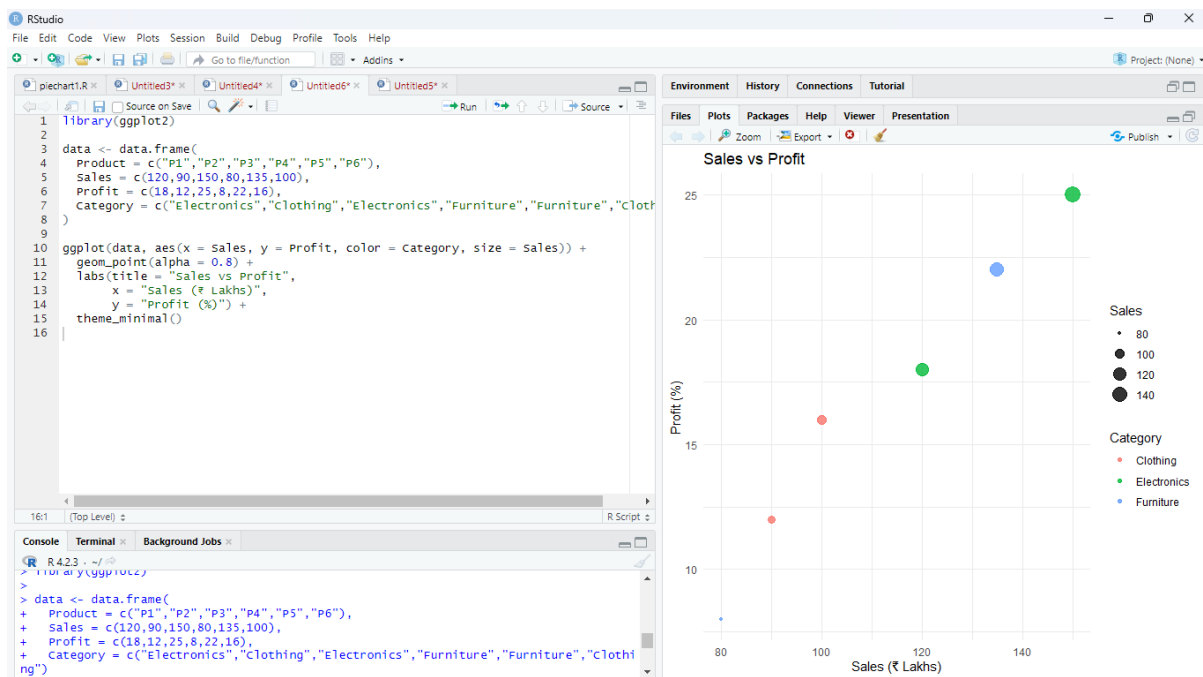
```
  labs(title = "Sales vs Profit",
```

```
        x = "Sales (₹ Lakhs)",
```

```
        y = "Profit (%)") +
```

```
  theme_minimal()
```

Output:



Question 2 (Climate Science)

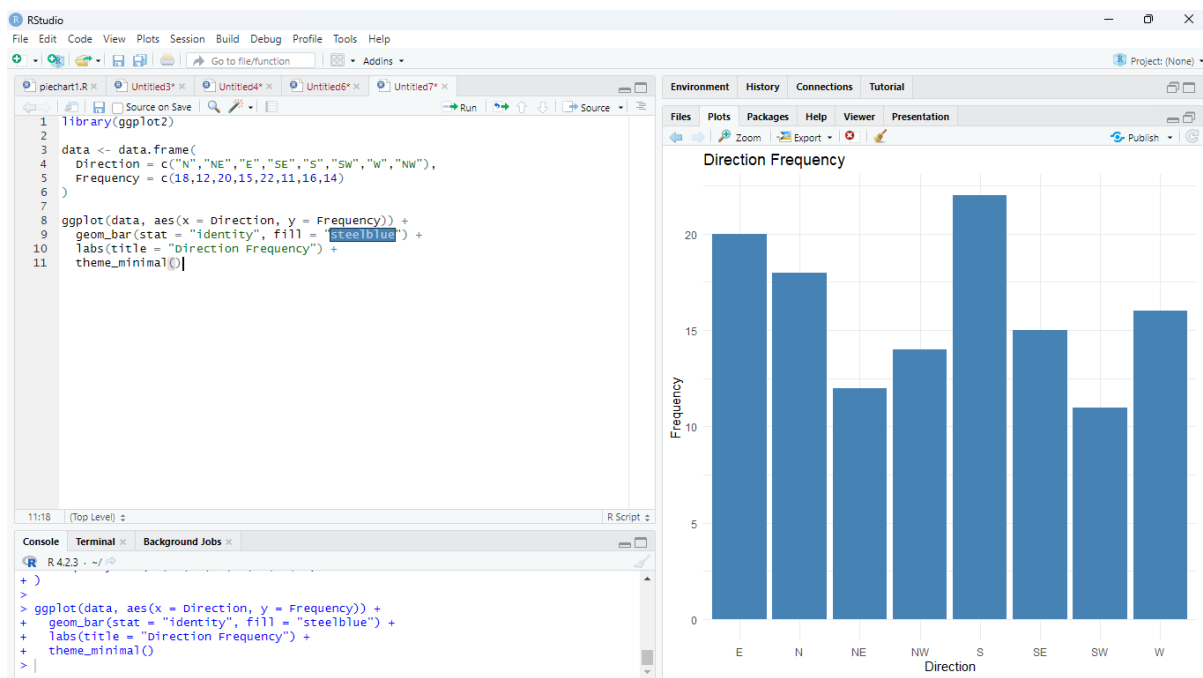
Program:

```
library(ggplot2)

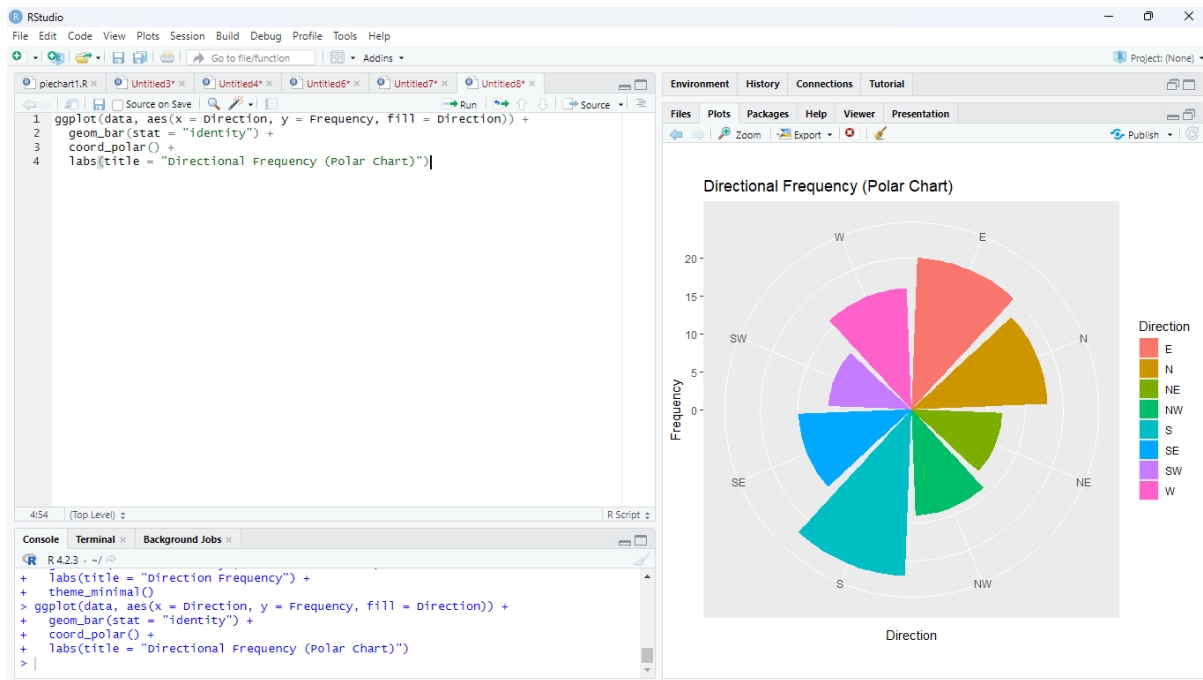
data <- data.frame(
  Direction = c("N","NE","E","SE","S","SW","W","NW"),
  Frequency = c(18,12,20,15,22,11,16,14)
)
```

```
ggplot(data, aes(x = Direction, y = Frequency)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Direction Frequency") +
  theme_minimal()
```

Output:



```
ggplot(data, aes(x = Direction, y = Frequency, fill = Direction)) +
  geom_bar(stat = "identity") +
  coord_polar() +
  labs(title = "Directional Frequency (Polar Chart)")
```



Question 3 (Business Analytics & Marketing)

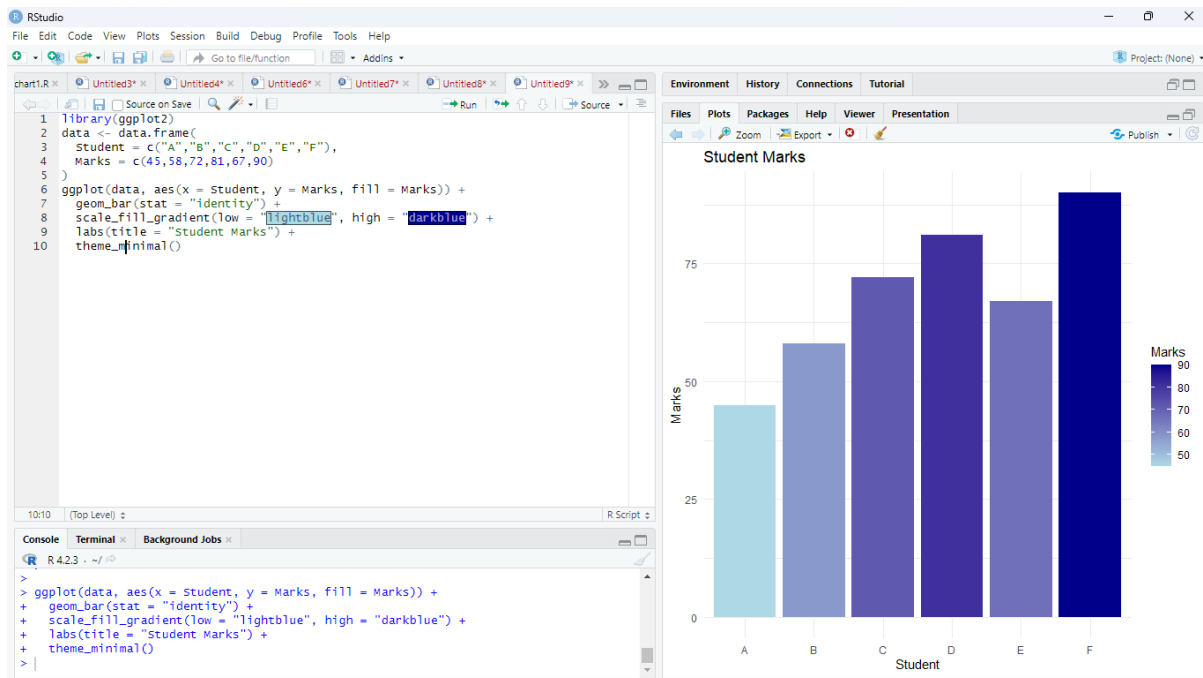
Program:

```
library(ggplot2)

data <- data.frame(
  Student = c("A", "B", "C", "D", "E", "F"),
  Marks = c(45, 58, 72, 81, 67, 90)
)

ggplot(data, aes(x = Student, y = Marks, fill = Marks)) +
  geom_bar(stat = "identity") +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Student Marks") +
  theme_minimal()
```

Output:



Question 4 (Transportation Engineering)

Program:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Region = c("North","South","East","West","Central"),
```

```
  Revenue = c(820,760,790,1010,770)
```

```
)
```

```
data$Highlight <- ifelse(data$Revenue == max(data$Revenue), "Highest", "Others")
```

```
ggplot(data, aes(x = Region, y = Revenue, fill = Highlight)) +
```

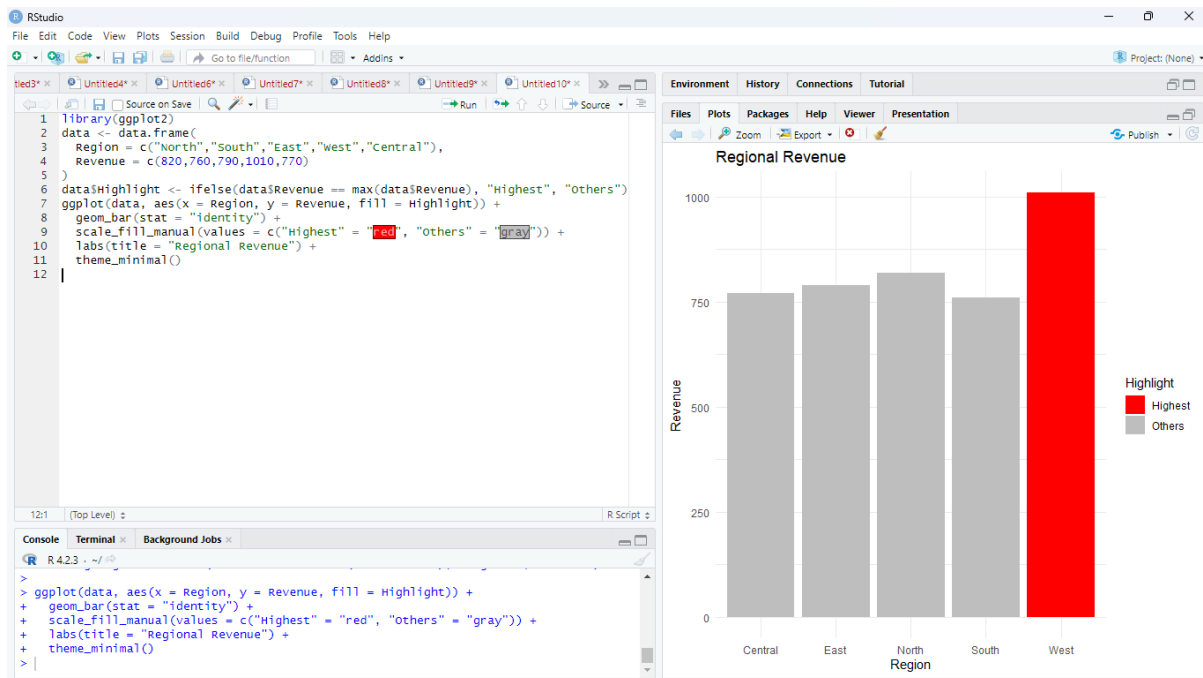
```
  geom_bar(stat = "identity") +
```

```
  scale_fill_manual(values = c("Highest" = "red", "Others" = "gray")) +
```

```
  labs(title = "Regional Revenue") +
```

```
  theme_minimal()
```

Output:



Question 5 (Environmental Sustainability Dashboard)

Program:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  City = c("A", "B", "C", "D", "E"),
```

```
  Population = c(8.2, 5.6, 9.4, 6.8, 4.9),
```

```
  AQI = c(110, 180, 90, 240, 150),
```

```
  Rainfall = c(850, 430, 1200, 520, 680),
```

```
  Region = c("North", "South", "North", "East", "West")
```

```
)
```

```
ggplot(data,
```

```
  aes(x = Population,
```

```
      y = AQI,
```

```
      size = Rainfall,
```

```
      color = Region)) +
```

```
geom_point(alpha = 0.8) +
```

```
labs(title = "Population vs AQI",
```

```
      x = "Population (Million)",
```

y = "AQI") +

theme_minimal()

Output:

